

Unit 12, Lesson 2: Finding an Unknown Fraction in an Equation

Benchmark

MA.6.AR.2.4 Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position.

Learning Target

- ☐ I can determine an unknown fraction in a numeric equation with three numbers.

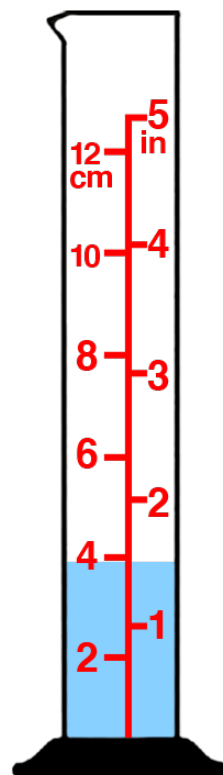
12.2.1 Warm-Up: How Much Rain?

José checked his rain gauge before and after a rain shower.

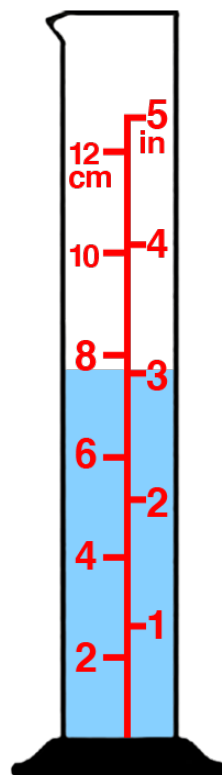
1. How much rain fell during the rain shower?

2. Use the pictures to describe how you determined the amount of rain that fell during the rain shower.

BEFORE



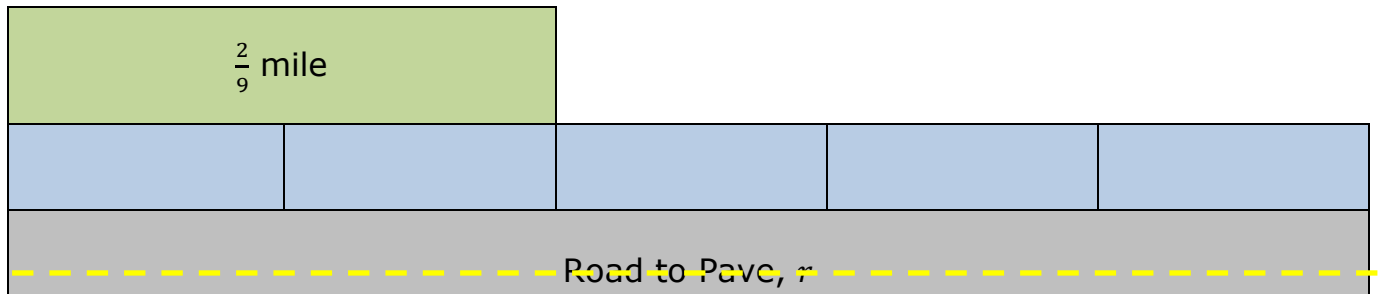
AFTER



12.2.2 Guided Instruction: Using Diagrams to Solve Equations with Fractions

Diagrams are helpful tools when solving equations for unknown values.

1. A road construction crew paved $\frac{2}{9}$ mile, which is $\frac{2}{5}$ of a full paving job. The bar model represents the equation $\frac{2}{9} = \frac{2}{5}r$, where r is the length of the whole road being paved.



- a. Discuss with your partner how the diagram represents the information given.
- b. Describe how to use the bar model to solve the equation for r , the length of the road being paved.
- c. How long is the road being paved?
- d. Check your solution. $\frac{2}{9} = \frac{2}{5} \left(\frac{\boxed{}}{} \right)$

2. Shirlene had $4\frac{1}{8}$ yards of ribbon. She decided to use some for her craft project. She has $2\frac{3}{4}$ yards of ribbon left.

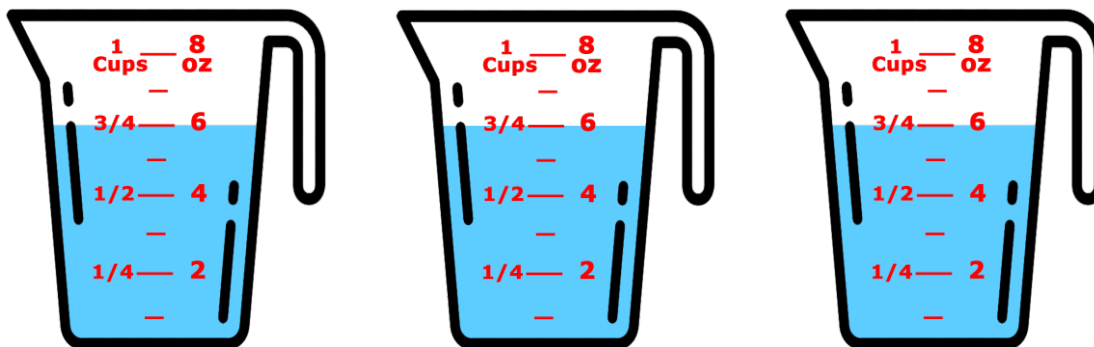
The model represents the amount of ribbon Shirlene had before she started her craft project.



The equation $4\frac{1}{8} - x = 2\frac{3}{4}$ can be used to find x , the amount of ribbon used for the craft project.

- Shade the part of the model that represents the amount of ribbon Shirlene has left after her craft project.
- What does the unshaded part of the model now represent?
- Use the model to find the value of x in the equation $4\frac{1}{8} - x = 2\frac{3}{4}$.
- How much ribbon did she use for the craft project?
- Check your solution. $4\frac{1}{8} - \boxed{} = 2\frac{3}{4}$

3. Donté used milk for 3 batches of a recipe. He used $\frac{3}{4}$ cup of milk for each batch.



The equation $m \div 3 = \frac{3}{4}$ can be used to find m , the amount of milk Donté used.

- a. Use the measuring cups shown to solve the equation $m \div 3 = \frac{3}{4}$.
- b. Check your solution. $\div 3 = \frac{3}{4}$

12.2.3 Collaboration: Solving Equations with Fractions Using Reasoning and Mental Math

1. Solve each equation. Check each of your solutions.

$$\frac{2}{3} \cdot d = \frac{10}{9}$$

$$1 = h + \frac{3}{7}$$

$$e \div \frac{5}{3} = 1$$

$$\frac{1}{4} f = \frac{1}{7}$$

$$\frac{1}{2} = 3\frac{1}{4} - w$$

$$x - 1\frac{1}{2} = 1\frac{1}{4}$$

2. Some of the equations have equivalent solutions. Ask two classmates for a pair of equations that have equivalent solutions. In the table, record their solution and write the equations they tell you gave them that solution. *You are the only person who should write in the first two columns of the table.* Have the person initial next to the equations they told you have equivalent solutions.

Solution	Equations	Initials

12.2.4 Guided Practice: Solving Equations with Fractions

1. Raul made jam and filled four equal-sized jars with $2\frac{2}{3}$ cups of jam. The equation $j \div 4 = 2\frac{2}{3}$ can be used to find the amount of jam, j , Raul made.
 - a. Draw a diagram that represents this situation.
 - b. How much jam did Raul make? Include units.
2. The equation $2 = \frac{11}{9} + n$ can be solved using reasoning.
 - a. How many ninths are equal to 2?
 - b. Explain how the answer to the previous question can be used to reason about the value of n .
 - c. What is the value of n ?

12.2.5 Your Turn!

1. Delia had a piece of decorative border. She cut $\frac{5}{6}$ feet off to make the piece $3\frac{1}{4}$ feet. The situation is modeled by the equation $t - \frac{5}{6} = 3\frac{1}{4}$, where t is the length in feet of the piece of trim before it was cut.
 - a. Draw a diagram that represents this situation.
 - b. How long was the piece of border before Delia cut it? Include units.

1. Mr. Bing placed a new 5-gallon water bottle in the office water dispenser when he arrived in the morning. At the end of the day, the bottle had $2\frac{7}{8}$ gallons remaining. The equation $5 = x + 2\frac{7}{8}$ can be used to find x , the gallons of water dispensed from the bottle.

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